Headline XVI: random phases and amplitudes in general dimension

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Abstract

Unit 212 completes programme A2 (Astra #22). For random inputs $X_n : \Omega \to C([0,1]^d)^2$ (phase, amplitude) converging in distribution to Z along a countably generated filter, and deterministic scales $N_n \to \infty$, the normalised chart integrals converge in distribution to the limit functional of the limiting input:

$$F_{N_n}(X_n) \;\Rightarrow\; F(Z), \qquad F_N = \frac{\int \eta\, u^h e^{-\beta N^2 u^{2k} + \beta N u^k \xi}}{N^{-p}(\log N)^{m-1}}, \quad F = \text{phaseCoeff}$$

(`tendstoInDistribution_normChart`). Proof: F is continuous (Lipschitz on balls), so $F(X_n) \Rightarrow F(Z)$ by continuous mapping; and $F_{N_n}(X_n) - F(X_n) \to 0$ in probability, because the limit law is tight in the Polish space $C([0,1]^d)^2$ (Mathlib's `isTightMeasureSet_singleton`), portmanteau puts most of X_n in the open δ-thickening of a compact K carrying most of Z, and on that thickening the uniform convergence on K (unit 211) extends by the Lipschitz bounds (units 209–210). Mathlib's Slutsky lemma `TendstoInDistribution.add_of_tendstoInMeasure_const` finishes. Zero `sorry`/`axiom`; 9089 jobs.

1 The things formalised

```
instance : MeasurableSpace (InputSpace d) := borel _;  instance : BorelSpace
      (InputSpace d)
theorem normChart_lipschitzOn (h ) (hN : 0   N) (R) :   L, 0   L    x y
      closedBall 0 R,
    |normChart h k l   N x - normChart h k l   N y|   L * dist x y
theorem continuous_normChart (h ) (hN : 0   N) : Continuous (normChart h k l   N)
theorem continuous_limChart (hk) (hl) (h ) (hmin) : Continuous (limChart h k l  )
theorem tendstoInMeasure_normChart_sub … (hZm : Measurable Z) (hX :
      TendstoInDistribution X L Z (fun _ =>  )  ')
    (hN : Tendsto Nseq L atTop) :
    TendstoInMeasure   (fun n   => normChart h k l   (Nseq n) (X n  ) - limChart h
      k l   (X n  )) L (fun _ => 0)
theorem tendstoInDistribution_normChart … (hXm :   n, Measurable (X n)) (hZm) (hX)
      (hN) (hN0 :   n, 0   Nseq n) :
    TendstoInDistribution (fun n   => normChart h k l   (Nseq n) (X n  )) L
      (fun   => limChart h k l   (Z  )) (fun _ =>  )  '
```

2 Mathematics

The key point, anticipated in Astra #21/#22, is that no tightness of the *sequence* X_n is needed: only the limit law is tight (automatic on a Polish space), and the passage from the compact K to its open δ-thickening U uses (i) portmanteau for the closed set U^c ($\limsup \mu(X_n \in U^c) \le \mu'(Z \in U^c) \le \mu'(Z \notin K)$), and (ii) the equi-Lipschitz bounds on the ball of radius $R+1 \supseteq U$: for $x \in U$ with $d(x,y) < \delta$, $y \in K$, $|F_N(x) - F(x)| \le L\delta + \sup_K |F_N - F| + L'\delta < \varepsilon$ once $(L+L')\delta \le \varepsilon/3$ and N is large. The random inputs enter only through their sup-norm topology; the deterministic Headline XIII does the analysis.

3 Lean notes

- `SecondCountableTopology C(K, )` needs the import `Mathlib.Topology.ContinuousMap.SecondCountableSpace` (instance `ContinuousMap.instSecondCountableTopology`, requiring a second countable locally compact domain); it was not in the import closure, and `#synth` fails silently otherwise. `isTightMeasureSet_singleton` needs `Mathlib.MeasureTheory.Measure.Tight`.
- `set S := leadScale …` folds only existing occurrences; a later `unfold normChart` reintroduces `leadScale`, so `rw [← hS]` again before case-splitting on `S = 0` (division by a possibly zero scale is handled by `div_zero`, no positivity needed).
- The portmanteau pattern from `NormBoundedInProbability` (`ProbabilityMeasure.limsup_measure_closed_le_of_tendsto`, `Measure.map_apply_of_aemeasurable`, `eventually_lt_of_limsup_lt`) transfers verbatim to the function space.
- A section variable `[L.IsCountablyGenerated]` that a theorem does not use triggers the linter; `omit [L.IsCountablyGenerated] in` before the theorem.

4 Status

A2 complete (units 209–212). Natural corollary: the random per-stratum posterior quotient $Z_N[\varphi c](X_n)/Z_N[c](X_n) \Rightarrow F^{\varphi c}(Z)/F^c(Z)$ via the existing positive-denominator quotient theorem (unit 213). Then review v12, report v3 / hand-off.